# Correlation-Based Feature Selection for Association Rule Mining in Semantic Annotation of Mammographic Medical Images

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**Abstract.** Mining of high dimension data for mammogram image classification is highly challenging. Feature reduction using subset selection plays enormous significance in the field of image mining to reduce the complexity of image mining process. This paper aims at investigating an improved image mining technique to enhance the automatic and semi-automatic semantic image annotation of mammography images using multivariate filters, which is the Correlation-based Feature Selection (CFS). This feature selection method is then applied onto two association rules mining methods, the Apriori and a modified genetic association rule mining technique, the GARM, to classify mammography images into their pathological labels. The findings show that the classification accuracy is improved with the use of CFS in both Apriori and GARM mining techniques.

**Keywords:** Correlation-based feature selection, multivariate filters, association rule mining, mammographic image classification, and semantic annotation.

## 1 Introduction

The advancement in medical imaging has resulted in many breakthroughs in the medical field. The use of digital images is essential in the diagnosis and also research, especially by the radiographers. As huge number of digital images is being created, it becomes problematic for storage and retrieval. In addition, the semantic gap problem, which is the difference in the representation of digital images and the actual keyword query from the user, increases the difficulty for effective retrieval of those medical images. Furthermore, the use of image as a query such as in the case of query by example (QBE) to overcome the semantic gap problem may not be practical in this context.

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Fortunately, recent technology in medical imaging allows for metadata being associated with the images. Those metadata can facilitate for text-based retrieval, in which those images are indexed based on their metadata or annotations and the keyword-based query submitted by the user will be matched to the indexed metadata for retrieval. Due to limited information stored as metadata for each image, such a system will not be effective for robust retrieval of medical images. Therefore, the performance of such a retrieval system can be significantly improved by adding semantic annotation onto the images for indexing.

Automatic and semi-automatic image annotation can be accomplished through classification of images according to label of concepts. Such labels will be considered as potential query terms, and are used to annotate those images for indexing. Fortunately, the set of label of concepts has been established in the medical field for manual indexing of resources, such as ICD10 [1] etc. The labels usually represent the pathological description of certain diseases. As such, this paper investigates the problem of annotating mammography images into *normal*, *benign* or *malignant* classes that represent the stages of breast cancer.

Association rules (AR) mining gain popularity as the classification technique for semantic annotation. Generally, this approach will generate many rules based on the frequent itemset generated for all features. Instead, a condition is set such that the consequence of the rules must be the class categories, *normal*, *benign* or *malignant*. As such, there will be many rules generated for each of the classes. Similar to other data mining approaches, the performance of AR depends largely of the features used for the mining task. A huge number of features used may enhance the classification accuracy, but will affect the efficiency of AR through the generation of too many itemset as well as rules. The number of features can be reduced through the process of feature selection to improve the efficiency and to maintain or improve the accuracy of the classification. As such, this paper investigates the effect of using Correlation-based Feature Selection (CFS) for association rules mining in the context of mammography image classification.

The remaining of this paper is ordered as follows. Related work is presented in Section 2. In Section 3, a detailed discussion on Correlation-based Feature Selection approach is presented. The experimental setup and analysis is given in Section 4. Section 5 describes the conclusion and future work.

## 2 Related Work

The quality of decision for automatic diagnosis in the medical field depends on the quality data. The elimination of redundant features reduces the data size. Integrating feature selection reduces number of features, removes noisy or irrelevant data, thus speeding up the mining process. Hence, mining on a reduced set of data helps to make the association rule pattern to be discovered easily and to improve its predictive accuracy. There are two types of data reduction methods, which are wrapper and filter methods. Even though wrapper methods can produce better result, they are expensive for the large dimensional database. On the other hand filter method is computationally simple and fast and precedes the actual association rule generation process. Filter methods use some properties of the data to select the feature. An intrinsic property such as entropy has been used as a filter method for feature selection. In [2], Lei Yu *et al.* proposed a selection method using a correlation measure that identifies redundant features. Many popular search procedures like particle swarm optimization, sequential forward selection, sequential backward selection, genetic search, etc. have been proposed in many researches. Genetic Algorithms are effectively applied to a variety of problems like feature selection problems [3], data mining problems [4], scheduling problems [5], machine learning problems [6,7], multiple objective problems [8,9], and traveling salesman problems [10]. Since the univariate filters does not justify for interactions between features multivariate filter correlation based feature selection (CFS) can be used to overcome the drawback of the univariate filter for determining the best feature subset.

Numerous authors have proposed many algorithms in recent years for mining association rules such as Genetic Algorithm (GA). Manish Saggar *et al.* [11] have used the GA to optimize the rules generated by the Apriori algorithm. The rule based classification systems can even predict negative rules with the improvements applied to GAs. Ashish Ghosh et al. [12] solved the multi-objective rule mining problems by representing the rules as chromosomes using Michigan approach. Virendra Kumar et.al [13] extracted interesting association rules using an optimized GA using the measures like interestingness, completeness, support and confidence. Nikhil Jain et.al [14] used a genetic algorithm for the whole process of optimization of the rule set to find a reduction of Negative and Positive Association Rule Mining. Amy et.al [15] proposed Hybrid genetic algorithm for mining workflow best practices, using correlation measures instead of traditional support and confidence. Peter et.al [16] used a genetic algorithm for a structured method to find the unknown facts in large data sets. Jesmin Nahar et al. [17] proposed Association rule mining for the detection of sick and healthy heart disease factors, using the UCI Cleveland dataset, a biological database. Sufal et.al [18] proposed a method of utilizing linkage among feature selections using Multi-objective Genetic Algorithm for data quality mining. Dong Gyu Lee et al [19] used Genetic algorithm for generating association rules related to hypertension and diabetes in discovering medical knowledge for young adults with acute myocardial infarction. Ramesh Kumar et.al [20] proposed a novel genetic algorithm for the prioritization of association rules produced by the Apriori algorithm and tested for the four different data sets. Basheer et.al [21] proposed a multi-objective genetic algorithm for generating association rules in discovering interesting association rules. J. Malar Vizhi et al. [22] proposed a genetic algorithm using multi-objective evolutionary framework for the Primary - tumor dataset to generate high quality association rules. Tournament selection and multipoint crossover methods were used for a large number of attributes in the database for flexibility. The approach has been tested for only numerical and categorical valued attributes. Bettahally et.al [23] compared conventional mining algorithm, i.e. Apriori algorithm with the proposed genetic algorithm in local search for privacy preserving over distributed databases. In the Apriori algorithm population is formed in only single recursion, but in genetic algorithm population is formed in every new production. To overcome the disadvantages of Apriori algorithm, Genetic algorithm can be used for association rule mining. Above research gap helps to improve the Apriori algorithm as well as an existing Genetic algorithm, which can further be improved.

## 3 Correlation-Based Feature Selection (CFS)

In general, several features are usually used to describe the character of an object. The characteristic features that are used to classify the normal and abnormal lesions can be represented as mathematical descriptions. Feature extraction methodology analyses mammogram images to extract the most prominent features that represent various classes of the images. Unlike the complicated process of a human observer to identify a mass, the machine makes decisions with only limited features. In this paper, the statistical texture features such as contrast, coefficient, entropy, energy, homogeneity and a few other Haralick [24] and Soh [25] features that efficiently classify the benign and malignant mammograms are extracted with the distance between the pixel of interest and its neighbor equal to 1 and the angle of 0 .Let p(i,j) be the (i,j)th entry in a normalized Gray Level Co-occurrence Matrix (GLCM).μ and are the mean and standard deviation for the rows and columns of the matrix.

The representation and quality of data may have an effect on a given task in machine learning. Theoretically, the more discriminating power can be achieved by having more features, but current machine learning toolkits are not sufficient to deal with up to date datasets [26]. Many of the features extracted during the training phase are either partially or completely irrelevant to an object that has no effect on the target concept. The feature selection process helps to remove redundant and irrelevant features, thus reducing the feature space. This minimizes the computation time and helps in improving the prediction accuracy [27]. Data reduction methods are of two types, wrapper and filter methods. On the other hand filter approach measures the feature subset relevance and is independent of learning algorithm. Since the univariate filters does not justify for interactions between features we use multivariate filter correlation based feature selection (CFS) to determine the best feature subset. Correlation based Feature Selection involves heuristic search to evaluate the subset of features. This is a simple filter algorithm based on correlation heuristic function. This function evaluates the subsets that are highly correlated and uncorrelated with each other and class. The features are accepted depending on its level of extent in predicting its class. The features that did not influence the class will be ignored as irrelevant features. The stages of CFS are shown in combination with data discretization in Fig. 1.

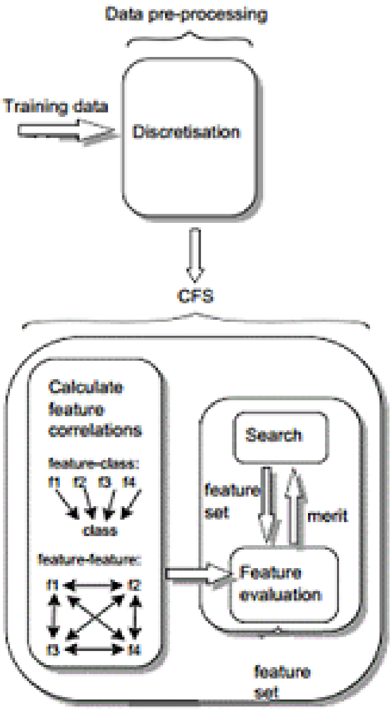



**Fig. 1.** The components of CFS

The mammogram training data are discretized first and then passed to CFS. For the prediction of class labels, this CFS considers the efficacy of individual features and its inter-correlation. If given the inter-correlation between each pair of features and the correlation between each of the features and its class, then the correlation between a subset of features selected for evaluation can be predicted from the formula

$$cr_{zc} = \frac{f\,\overline{cr_{zi}}}{\sqrt{f + f(f-1)\overline{cr_{ii}}}}$$

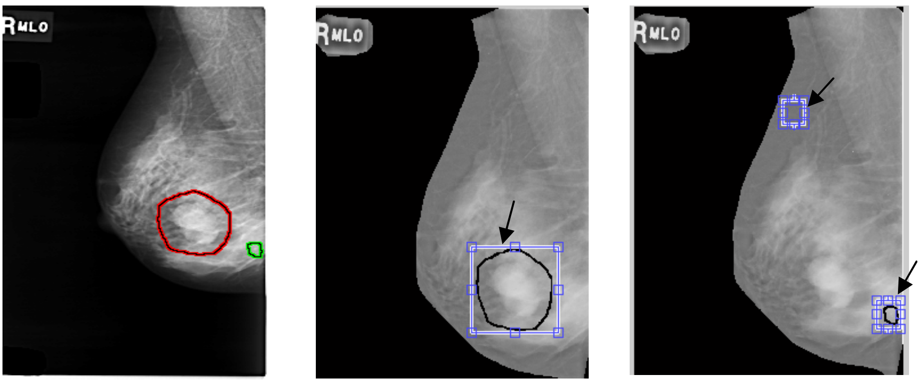
where $cr_{zc}$ = heuristic merit of a feature subset for $f$ number of features, $c\,\overline{r_{zi}}$ the = average of the correlations between the class and the features, $c\,\overline{r_{ii}}$ = average inter-correlation between feature pairs [28]. The subset with the highest $cr_{zc}$ value is used to reduce the data dimensionality.

The feature selection process together with CFS undergoes some search procedure. This paper uses Weka GA as a search method with CFS as a subset evaluating mechanism. Genetic algorithm (GAs) is modeled based on the process of natural selection. At every iteration new populations are generated from old ones in each iteration. They are actually binary encoded strings. Every string is evaluated to measure its fitness value for the problem. Likewise the entire generation of new strings can be computed using the genetic operators, on an initially random population. This operative way of discovering large search space is essential for feature selection. The individual fitness can be decided by the correlation between the features. Based on the correlation coefficient the individuals will be assigned a rank by the fitness function. The features that have the lower correlation coefficients and with higher fitness value will be appropriate for crossover operations.

### 3.1 Dataset and Selection of ROI

The data set used in this experiment is taken from the digital database for mammography from the University of South Florida [29], which is DDSM. All images are digitized using LUMISYS Scanner at a resolution of 50 microns, and at 12 bit grayscale levels. The dataset consists of 240 images that include three categories of which 80 are normal, 80 are benign and 80 are malignant. Then, the region of interest (ROI) is isolated within those images as the preprocessing step. We use the contour supplied together with the images in the DDSM dataset to extract ROIs of size 256 x 256 pixels.

A total of 242 ROIs are extracted with the mass centered in a window of size 256×256 pixels, where 162 are abnormal ROIs (circumscribed masses, speculated masses, ill-defined masses and architectural distortion) and 80 are normal ROIs.




(a) (b) (c)

**Fig. 2.** Example of ROI for (a) original image, abnormal (b) and normal cases (c)

### 3.2 Association Rules Mining

For each image, the selected discretized features are stored in such a way that *n* columns represent *n* features while the last column represent a class (eg. *normal, benign* and *malignant*). Two association rules mining techniques are used to discover the frequent itemset and eventually to generate the rules, which are the classical Apriori algorithm and also a modified genetic algorithm based association rule mining, GARM. The Apriori algorithm tends to produce a huge number of rules, that is redundant and may not be efficient. On the other hand, genetic algorithm (GA), such as

GARM can generate good rules by performing a global search with better attribute interactions thus can improve the effectiveness of association rule mining. For each category in the database, the GA is applied separately to construct sets of rules. For each rule the fitness value is calculated and the rule that has the highest fitness value in each population will be stored as the global best rule. Then, the best rules from each category are pooled to form rule set. New population can be computed using the genetic operators like reproduction, crossover, and mutation to extract the best local rules.

### 3.3 Classification

The extracted set of rules represents the actual classifier. It classifies a new image to its category. When a new image is provided, a feature vector is extracted and it searches in the rules for matching classes. Then, the number of matched rules for each class is calculated. Based on the matched rules in each class, the average *confidence* score is calculated. The class for the new image is identified based on the highest average *confidence* score in the class and the number of rules matched. The algorithm describes the classification of a new image.

***Input:*** Number of category *C*, list of training rules for each category $C_i$, Number of rules **n** in each category $C_j$, Total number of rules *N*

***Output:*** Category attached to the new image

*for each category Ci*
*for each rule R do*
*If R matches I then*
**MC++**
else
**NMC++**

end if
*end for*
*end for*

*for each category Ci subset*
find P= $\frac{MC}{n}$ the percentage of set of rules that match I
find average confidence Q of rules
*end for*

Put the new image in the category Ci that has the highest average confidence and higher percentage of matching

### 3.4 Experimental Results

In order to evaluate the performance of the CFS feature selection method, the 10-fold cross validation is used. Ten equal groups with an equal number of images from each class is formed. One group is selected as the test images and the remaining nine groups are used to generate the rules. In the classification stage, the class for each image in the test group is identified and average accuracy measure (AC) of the classification is measured as follows:

$$AC= [(nc_1+ nc_2 +nc_3)/N_T] * 100 \quad (1)$$

where $nc_1$, $nc_2$ and $nc_3$ are the number of correctly classified images in class $C_1$, $C_2$ and $C_3$, respectively, while $N_T$ is the total number of images in the test group.

**Table 1.** The average accuracy measures (AC) for CFS with different AR algorithms

| Group | Apriori | | GARM | |
|---|---|---|---|---|
| | **w/o CFS** | **with CFS** | **w/o CFS** | **with CFS** |
| 1 | 79.9 | 82.6 | 81.7 | 87.9 |
| 2 | 83.4 | 85.7 | 84.4 | 89.4 |
| 3 | 78.6 | 81.8 | 80.9 | 88.6 |
| 4 | 85.2 | 85.6 | 86.8 | 92.6 |
| 5 | 82.8 | 83.6 | 82.1 | 85.4 |
| 6 | 83.9 | 86.9 | 84.9 | 87.2 |
| 7 | 79.6 | 83.2 | 79.7 | 87.9 |
| 8 | 80.4 | 82.8 | 83.5 | 86.3 |
| 9 | 82.5 | 84.7 | 82.3 | 85.7 |
| 10 | 84.3 | 87.4 | 85.1 | 90.2 |
| **Average AC** | **82.06** | **84.43 (+2.9%)** | **83.13** | **88.12 (+6%)** |

The average accuracy measure (AC) for CFS with Apriori and GARM algorithms is a depicted in Table 1. The result shows that a marginal performance improvement can be achieved when CFS is used before the application of both AR algorithms, with an improvement of 2.9% and 6% for the average AC on Apriori and GARM algorithm, respectively. This finding indicates that the feature selection process can potentially improve the effectiveness of AR based classification on mammography images. It is also learnt that CFS can have a bigger impact on the performance of genetic algorithm based association rules mining as compared to the classical Apriori algorithm.

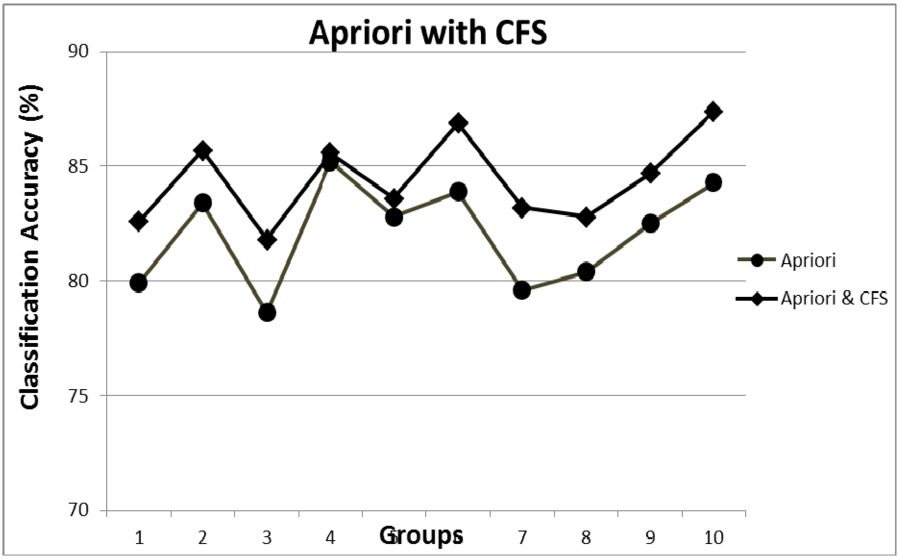



**Fig. 3.** Comparison of AC for Apriori algorithm based on group

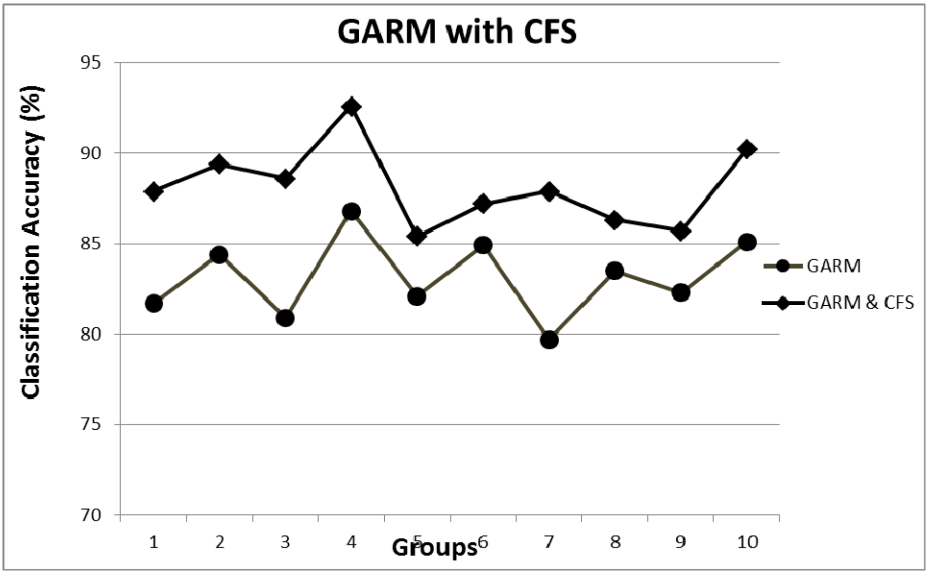



**Fig. 4.** Comparison of AC for GARM algorithm based on group

Fig. 3 & Fig. 4 show the comparison of AC for different algorithms in each group. Based on the figure, it clearly shows that the classification has more effective as the CFS consistently shown better performance in all groups as compared to not using CFS for both AR algorithms. An interesting findings show that genetic algorithm based association rules without CFS does not have a significant impact on the classification performance as compared to the classical Apriori algorithm without CFS which is only 1.7% difference, despite being more efficient. It also learnt that the difference in performance for both classical Apriori algorithm and genetic algorithm can be enlarged with the use of CFS with 4.4% difference.

## 4 Conclusion

This paper investigates the effect of using feature selection process, CFS, on two AR algorithms, the classical Apriori algorithm and a genetic algorithm based AR algorithm, GARM, for the mammography image classification problem. The result is promising, where the algorithms are able to achieve 88% correct classification using GARM with CFS compared to [30] without any feature selection. In addition, the use of feature selection, CFS, can improve the classification performance of both AR algorithms. However, the benefit of using an efficient GARM algorithm does not necessarily improve the classification accuracy. Our further work will be on GA for optimization of Association Rule Mining to improve the classification accuracy for an automatic diagnosis and suggestions of possible pathological terms.

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